TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

Reminders
Words: at least ≥ · at most ≤ · more than > · fewer than <.
Solve like an equation: undo + and −, then × and ÷.
Flip the sign when you multiply or divide both sides by a negative number.
Graph: ○ for < and >, ● for ≤ and ≥. Shade right for > and ≥, left for < and ≤.

1. What is the solution set of $-4x > 12$?

Tip: ÷ by a negative → flip the sign

A. $x > -3$

B. $x < 3$

C. $\boldsymbol{x < -3}$

Divide by −4 and flip: $x < -3$.

2. What is the solution set of $2x + 3 \leq x - 1$?

Tip: get [[x]] on one side, like an equation

F. $\boldsymbol{x \leq -4}$

G. $x \geq -4$

H. $x \leq -2$

Subtract x: $x + 3 \leq -1$. Subtract 3: $x \leq -4$.

3. Which inequality means the same as $-6 \leq x$?

> **Tip:** switch sides → the sign points the other way

A. $x \leq -6$

B. $x \geq -6$

C. $x > -6$

Read it from x's side: x is greater than or equal to -6.

4. What value of x makes $3(x - 2) = 9$ true?

> **Tip:** distribute, then undo + and ×

F. 3

G. 11/3

H. 5

$3x - 6 = 9$, $3x = 15$, $x = 5$.